Class-Imbalance Mitigation on TCGA-UT with a LoRA-Adapted Foundation-Model Encoder

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Abstract

On BRACS, six established mitigation methods recovered at most about half of the balanced accuracy lost under severe class imbalance with a Virchow2 encoder adapted through low-rank adaptation (LoRA). This report repeats that study on TCGA-UT, a 30-class dataset on which imbalance costs a frozen readout only 2.91 percentage points at a ratio of 100. The study uses 30 patient-disjoint split-draw shards and a balanced cross-entropy (CE) reference. CE and every mitigation method also train at imbalance ratios of 50 and 100 and at native class shares. Validation patients determine each method's strength per shard. Under LoRA, CE loses 1.56 points at ratio 100, 0.67 points at ratio 50, and 0.42 points at native shares: between a third and a half of the frozen readout's losses. At ratio 100, logit adjustment removes the loss: its train-time form recovers 1.54 points and its post-hoc form 1.41 points, and the intervals for both recovery shares include 1. Logit adjustment and classifier-only Gaussian clouded logit (GCL) adjustment are also the only methods that restore most of the rarest classes' recall. A curriculum of data augmentation (CUDA) recovers 1.02 points but also improves the balanced model by 0.81 points, so its gain does not depend on imbalance. At native shares, it is the only method that gains a full point. After subtracting their slightly negative balanced gains, end-to-end GCL, classifier-only GCL, and classifier retraining (cRT) recover 1.48, 1.40, and 1.10 points of imbalance-specific loss. At ratio 100, a prior correction at inference therefore recovers nearly all of the loss, and differences between methods remain below one point.

1 Motivation

With patient support per class fixed, class imbalance reduces balanced accuracy (BA) by 2.91 percentage points on TCGA-UT and 7.65 points on BRACS at an imbalance ratio of $\rho = 100$, using a logistic-regression readout on frozen Virchow2 features (Queisler, 2026e,d). On TCGA-UT, two sources of loss compound (Queisler, 2026a). In the *frequency* effect, common classes dominate the training loss and pull the decision boundary toward themselves. In the *support* effect, rare classes offer too few distinct patches to capture their variation. These effects alone cost 1.20 and 0.53 points, respectively, at $\rho = 100$; their interaction costs a further 1.18 points.

A companion study measured how much of BRACS's loss six established long-tailed methods recover when the Virchow2 encoder (Zimmermann et al., 2024) uses low-rank adaptation (LoRA) (Queisler, 2026b,c). There, cross-entropy (CE) training lost 4.95 points at $\rho = 100$. A curriculum of data augmentation (CUDA) recovered about half of that loss and logit adjustment about two-fifths; no method closed the gap. That study covered one dataset. TCGA-UT differs from BRACS in several respects relevant to mitigation: it has 30 classes instead of seven, a balanced accuracy near 72 instead of 37 percent, and an imbalance loss less than half as large. This report asks: *how much of TCGA-UT's imbalance loss does each method recover under a LoRA-adapted encoder, and do the conclusions drawn on BRACS carry over to a dataset on which imbalance costs little?*

2 Design

The shards follow the imbalance design of Queisler (2026e). The 30 TCGA-UT cancer types have three fixed patient-disjoint splits, each sampled in ten draws. Each draw samples $G = 20$ training patients per cancer type from those with at least 160 training patches. The total patch budget, $T = 30 \cdot G \cdot 32 = 19{,}200$, remains fixed across allocations. Within each of the 30 split-draw shards, all allocations use the same patients and the same random assignment of classes to allocation ranks; only the class shares differ. Validation and test partitions retain their natural class distribution. Training uses decoded patch images because every method adapts the encoder through a rank-8 LoRA update on the attention projections, with a linear head (a cosine head for Gaussian clouded logit, GCL, adjustment), as defined by Queisler (2026c).

2.1 Allocations

Each shard's patch budget is divided among the 30 classes in four ways, following an exponential allocation profile. The *balanced* allocation ($\rho = 1$) trains CE as the *undamaged reference*, along with methods whose effect on balanced data is not zero by construction (Section 3). Three imbalanced allocations use ratios $\rho = 50$ and $\rho = 100$ and TCGA-UT's native class shares (N, realized ratio about 18.1). Each trains CE as its *damaged reference* and every mitigation method over the grid in Table 1. At $\rho = 100$, the largest class receives 2,843 patches and the rarest 28, between one and two per patient. This ratio approaches the most severe imbalance the design can express without removing patients from the tail.

Methods fall into two training families. *End-to-end* methods (balanced sampling, CUDA, train-time logit adjustment, GCL) train the LoRA update and head together. CUDA (Ahn et al., 2023) augments image patches before the encoder, with a per-class strength that rises as the model learns the class, so it acts only end to end. *Classifier-only* methods freeze an end-to-end model's encoder and adjust only classifier parameters. Post-hoc logit adjustment, distribution alignment (DisAlign), and classifier re-training (cRT) start from the allocation's CE model. GCL's classifier-only form, its second training stage, starts from the end-to-end GCL model with the same σ. Logit adjustment and GCL therefore each appear in both families, giving eight method variants.

Training	Method	Strength	Grid
End to end	CE (reference)	–	–
End to end	Balanced sampling	s	0.5, 1.0
End to end	CUDA	p_{aug}	0.25, 0.5, 1.0
End to end	Logit adjustment (train-time)	τ	0.25, 0.5, 1.0
End to end	GCL	σ	0.02, 0.05, 0.2 (N: also 0.5)
Classifier only	Logit adjustment (post hoc)	τ	0.25, 0.5, 1.0, 1.5, 2.0
Classifier only	DisAlign	γ	0.5, 1.0, 1.5, 2.0 (N: also 3.0)
Classifier only	cRT	–	–
Classifier only	GCL	σ of its source	as GCL

Table 1: Strength grids for the imbalanced allocations. CUDA (curriculum of data augmentation) uses a fixed acceptance threshold (0.6), check size (10), and maximum strength (10). The balanced allocation trains cross-entropy (CE), CUDA over its grid, both Gaussian clouded logit (GCL) forms at $\sigma = 0.05$, distribution alignment (DisAlign) at $\gamma = 1$, and classifier re-training (cRT). N denotes native class shares.

2.2 Training length

End-to-end training length was fixed in advance using CE alone. CE models trained on the balanced and $\rho = 100$ allocations, with one draw per split, for 300, 600, 1,200, and 2,400 steps. The selection rule chose the smallest step count within 0.5 points of the best mean validation BA at $\rho = 100$. The four candidates reached 59.9, 65.3, 69.1, and 69.8 percent, respectively. All end-to-end fits therefore used 2,400 steps (batch size 256, learning rate 10^{-4}, cosine schedule), about 32 passes over the 19,200 training patches; classifier-only fits used 200 steps. Training length remains fixed across allocations within each family, so it cannot explain differences in recovery between allocations. Per shard, the grid contains 42 end-to-end and 44 classifier-only configurations, giving 1,260 and 1,320 fits across the 30 shards. Every configured fit completed and entered the analysis.

3 Evaluation

The primary endpoint is test patient-macro BA: the mean of per-class patch recalls, with equal weight for every test patient within a class. Estimates pool the 30 shards with equal weight per split. Across 2,000 replicates, uncertainty combines Bayesian bootstrap weights on test patients (Rubin, 1981) with resampling of draws within each split. All allocations and methods share the replicate weights, so every contrast is paired. Reported intervals are 95 percent percentile intervals.

For each method, shard, and imbalanced allocation, the grid value with the best *validation* BA is selected; selection never uses the test partition. With m^* denoting the selected fit and a the allocation,

$$D_a = \mathrm{BA}(\mathrm{CE}, 1) - \mathrm{BA}(\mathrm{CE}, a), \qquad R_m = \mathrm{BA}(m^*, a) - \mathrm{BA}(\mathrm{CE}, a), \qquad \mathrm{share}_m = R_m / D_a, \quad (1)$$

where the share is computed per replicate. Recovery counts as *material* if R_m is at least one point and its interval lies above zero. It counts as *full* if the interval for $\mathrm{BA}(m^*, a)$ contains the point estimate of $\mathrm{BA}(\mathrm{CE}, 1)$. Because that interval is unpaired, the paired share better indicates how much damage remains.

R_m compares a method with CE, combining imbalance correction with any general effect on accuracy. To separate these effects, the methods also train on the balanced allocation with the same validation selection. A method's *balanced gain* and *imbalance-specific recovery* are

$$G_m = \mathrm{BA}(m^*, 1) - \mathrm{BA}(\mathrm{CE}, 1), \qquad R_m - G_m = D_a - \big[\mathrm{BA}(m^*, 1) - \mathrm{BA}(m^*, a)\big], \quad (2)$$

with the latter measuring the difference between the damage to CE and the damage to the method. At equal class counts, balanced sampling and both forms of logit adjustment are identical to CE, so $G_m = 0$ by construction and they need no balanced fits. GCL's cloud size and DisAlign's reweighting are also uniform under balanced allocation. Their strengths therefore have no effect, and each uses one value; GCL then reduces to CE with a cosine head.

Secondary analyses are descriptive: head, body, and tail recall by assigned class rank at $\rho = 100$; macro negative log-likelihood (NLL) and expected calibration error (ECE), raw and after validation temperature scaling; the frequency of each selected strength; and the mean recovery share for each training family. The frozen-readout damage (Queisler, 2026e) uses the same shards and test patients but a different classifier, so it serves as an external reference without a paired interval. Comparisons with BRACS (Queisler, 2026b) are likewise descriptive.

	$\rho = 50$ ($D = 0.67$)		$\rho = 100$ ($D = 1.56$)		N ($D = 0.42$)
Method	R_m	share	R_m	share	R_m
Logit adj. (train-time)	0.86 [0.53, 1.19]	1.27 [0.81, 2.25]	**1.54** [1.21, 1.87]	0.99 [0.80, 1.24]	0.54 [0.20, 0.88]
Logit adj. (post hoc)†	0.92 [0.75, 1.10]	1.37 [0.95, 2.55]	**1.41** [1.18, 1.65]	0.91 [0.75, 1.13]	0.35 [0.14, 0.54]
CUDA	0.84 [0.52, 1.16]	1.25 [0.74, 2.45]	**1.02** [0.70, 1.34]	0.65 [0.45, 0.89]	**1.01** [0.63, 1.41]
GCL	0.53 [0.11, 0.95]	0.78 [0.20, 1.57]	0.93 [0.59, 1.29]	0.60 [0.40, 0.81]	0.03 [−0.39, 0.45]
DisAlign†	0.68 [0.58, 0.79]	1.01 [0.69, 1.90]	0.83 [0.73, 0.92]	0.53 [0.43, 0.68]	0.42 [0.27, 0.56]
GCL†	0.07 [−0.41, 0.51]	0.10 [−0.93, 0.68]	0.80 [0.36, 1.24]	0.51 [0.26, 0.73]	−0.45 [−0.87, −0.01]
Balanced sampling	0.18 [−0.12, 0.50]	0.27 [−0.29, 0.68]	0.70 [0.40, 1.01]	0.45 [0.28, 0.62]	0.32 [0.01, 0.62]
cRT†	0.21 [0.04, 0.37]	0.32 [0.07, 0.62]	0.68 [0.45, 0.91]	0.44 [0.31, 0.59]	−0.21 [−0.39, −0.05]

Table 2: At $\rho = 100$, the recovery-share intervals for both forms of logit adjustment include 1; all others lie below it. Recovery R_m in balanced-accuracy (BA) points and share of cross-entropy (CE) damage, with paired 95% intervals; bold marks material recovery ($R_m \geq 1$, interval above zero). †Classifier only, frozen encoder; unmarked rows train end to end. Recovery shares under native allocation (N) are omitted because the damage is too small for a stable ratio. CUDA: curriculum of data augmentation; GCL: Gaussian clouded logit adjustment; cRT: classifier retraining; DisAlign: distribution alignment.

4 Results

4.1 Damage under LoRA

Imbalance costs the LoRA model between a third and a half of what it costs the frozen readout. The balanced CE model reaches 71.51 percent BA [70.25, 72.66]. It loses $D = 0.67$ points [0.34, 1.04] at $\rho = 50$, 1.56 points [1.17, 1.94] at $\rho = 100$, and 0.42 points [0.06, 0.79] at native shares. All three intervals exclude zero, but only the damage at $\rho = 100$ reaches one point. The frozen readout lost 1.85, 2.91, and 1.18 points on the same shards (Queisler, 2026e). Two differences account for the smaller loss under LoRA. Its balanced model is 0.55 points less accurate than the frozen readout (71.51 versus 72.06 percent), while its imbalanced models are more accurate (70.84 versus 70.21 at $\rho = 50$, 69.95 versus 69.14 at $\rho = 100$, and 71.09 versus 70.88 at N). On BRACS, only the first difference was present.

4.2 Recovery per method

Figure 1 and Table 2 show each method's paired recovery. At $\rho = 100$, three of eight variants achieve material recovery: train-time logit adjustment (1.54 points [1.21, 1.87]), post-hoc logit adjustment (1.41 [1.18, 1.65]), and CUDA (1.02 [0.70, 1.34]). Both forms of logit adjustment recover essentially all of the damage. Their recovery shares are 0.99 [0.80, 1.24] and 0.91 [0.75, 1.13], respectively. The train-time model reaches 71.49 percent BA, versus 71.51 for balanced CE. End-to-end GCL (0.93 points), DisAlign (0.83), classifier-only GCL (0.80), balanced sampling (0.70), and cRT (0.68) recover less than one point, with intervals above zero. Their recovery shares range from 0.44 to 0.60, with intervals below 1.

At $\rho = 50$, no method achieves material recovery: all observed recoveries remain below one point. Both forms of logit adjustment, CUDA, and DisAlign recover 0.68 to 0.92 points, matching the damage: their recovery shares range from 1.01 to 1.37, with intervals that include 1. These shares are imprecise because the denominator is small. At native shares, CUDA gains 1.01 points [0.63, 1.41], more than twice the damage, and reaches 72.10 percent BA, above balanced CE. The other recoveries remain below 0.6 points. cRT and classifier-only GCL lower accuracy by 0.21 and 0.45 points, with recovery intervals of [−0.39, −0.05] and [−0.87, −0.01], respectively. Every variant meets the full-recovery criterion at every allocation. This criterion does not discriminate here: the unpaired accuracy intervals span about 2.4 points, wider than any of the damage estimates.

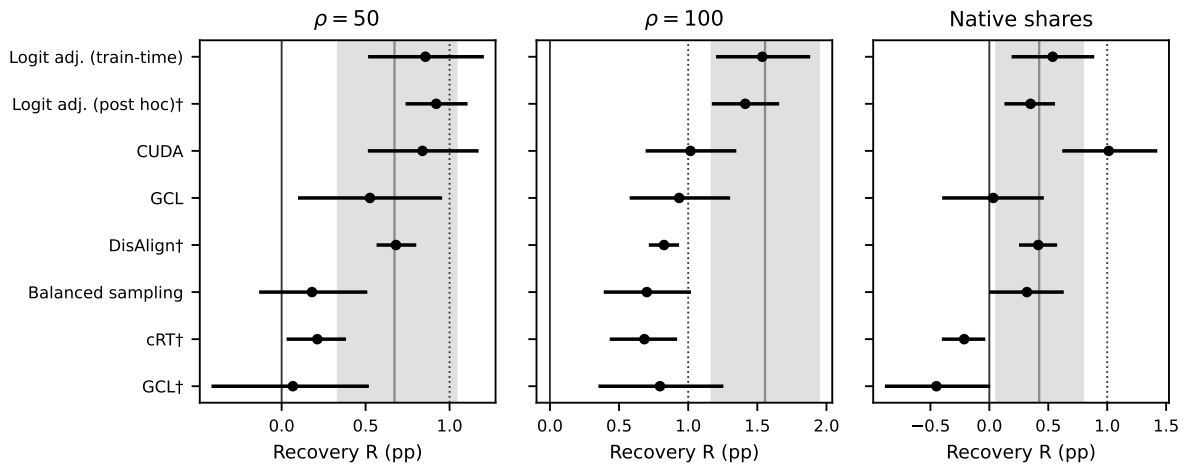


Figure 1: Logit adjustment recovers the full damage at $\rho = 50$ and $\rho = 100$; at native shares only CUDA (curriculum of data augmentation) gains a point. Paired recovery R_m in balanced-accuracy points with 95% intervals. The grey line and band mark cross-entropy (CE) damage D and its interval; the solid line marks zero and the dotted line the one-point threshold. GCL: Gaussian clouded logit adjustment; cRT: classifier re-training; DisAlign: distribution alignment; N: native class shares. †Classifier only.

4.3 Imbalance-specific recovery

Balanced gains differ in direction (Table 3). CUDA reaches 72.32 percent BA on the balanced allocation, a gain of $G = 0.81$ points $[0.48, 1.14]$ over CE. End-to-end GCL (-0.55 $[-0.87, -0.24]$), its classifier-only form (-0.60 $[-0.94, -0.27]$), and cRT (-0.41 $[-0.54, -0.28]$) lower BA on the balanced allocation, all with intervals below zero. DisAlign is indistinguishable from CE.

After subtracting its balanced gain, CUDA corrects none of the imbalance. Its imbalance-specific recovery is 0.20 points $[-0.17, 0.57]$ at $\rho = 100$, 0.03 at $\rho = 50$, and 0.20 at native shares, all with intervals that include zero. Methods with a negative balanced gain correct more than their recovery against CE suggests. At $\rho = 100$, end-to-end GCL corrects 1.48 points $[1.03, 1.96]$, a share of 0.95 $[0.72, 1.21]$; classifier-only GCL corrects 1.40 points $[0.87, 1.90]$, a share of 0.90 $[0.63, 1.13]$; and cRT corrects 1.10 points $[0.85, 1.34]$, a share of 0.71. All three achieve material imbalance-specific recovery. DisAlign's recovery remains 0.83 points. Five variants therefore meet the material-recovery criterion at $\rho = 100$ after subtracting balanced gains: both forms of logit adjustment, whose G_m is zero, both forms of GCL, and cRT. CUDA no longer meets the criterion.

4.4 End-to-end versus classifier-only training

The two matched pairs behave differently. Post-hoc logit adjustment, applied to the CE model at inference without retraining, recovers nearly as much as its train-time form (1.41 versus 1.54 points at $\rho = 100$, 0.92 versus 0.86 at $\rho = 50$). Re-fitting GCL's classifier recovers less than end-to-end GCL: recovery falls from 0.93 to 0.80 points at $\rho = 100$ and from 0.53 to 0.07 at $\rho = 50$. At $\rho = 100$, mean recovery shares are 0.67 for end-to-end and 0.60 for classifier-only methods, or 0.70 and 0.63 when both GCL variants are excluded. The family shares differ by 0.08 with GCL included and 0.07 with GCL excluded, calculated from unrounded estimates. These family and pair comparisons are point estimates without intervals.

Method	G_m	$R_m - G_m$			$\rho = 100$
		$\rho = 50$	$\rho = 100$	N	share
CUDA	0.81 [0.48, 1.14]	0.03 [−0.30, 0.37]	0.20 [−0.17, 0.57]	0.20 [−0.23, 0.63]	0.13 [−0.13, 0.34]
GCL	−0.55 [−0.87, −0.24]	**1.07** [0.57, 1.55]	**1.48** [1.03, 1.96]	0.58 [0.07, 1.07]	0.95 [0.72, 1.21]
GCL†	−0.60 [−0.94, −0.27]	0.67 [0.14, 1.20]	**1.40** [0.87, 1.90]	0.15 [−0.37, 0.66]	0.90 [0.63, 1.13]
cRT†	−0.41 [−0.54, −0.28]	0.63 [0.43, 0.82]	**1.10** [0.85, 1.34]	0.20 [0.00, 0.41]	0.71 [0.56, 0.92]
DisAlign†	0.00 [−0.01, 0.01]	0.68 [0.58, 0.79]	0.83 [0.73, 0.92]	0.42 [0.27, 0.56]	0.53 [0.43, 0.68]

Table 3: After subtracting its balanced gain, CUDA (curriculum of data augmentation) corrects none of the imbalance; both forms of Gaussian clouded logit (GCL) adjustment and classifier retraining (cRT) correct more than their recovery against cross-entropy (CE) suggests. Balanced gain G_m and imbalance-specific recovery $R_m - G_m$ (Eq. 2) in balanced-accuracy (BA) points, with the latter's share of CE damage at $\rho = 100$; paired 95% intervals, bold marks material recovery. †Classifier only. Balanced sampling and logit adjustment have $G_m = 0$ by construction, so their R_m in Table 2 is already imbalance-specific. DisAlign: distribution alignment; N: native class shares.

4.5 Recovery by class rank

At $\rho = 100$, CE retains 61.2 percent recall in the tail third of classes by assigned rank, versus 71.6 percent on the balanced allocation, and gains 6.3 points in the head (Figure 2). Logit adjustment restores most of the tail recall, reaching 69.4 percent (train-time) and 69.9 percent (post hoc), at a cost of about four points of head recall. End-to-end GCL and cRT raise tail recall to about 66 percent, and DisAlign and balanced sampling to about 65 percent. CUDA does not rebalance recall: it raises head and body recall by 0.7 points and tail recall by 1.7 points, leaving the latter at 62.9 percent. Classifier-only GCL raises tail recall to 73.3 percent, above the balanced model's 71.6 percent, while head recall falls to 68.7 percent, versus 71.3 percent on the balanced allocation. Rank-group recalls are point estimates.

4.6 Selected strengths

Validation selection favours mild corrections. At $\rho = 100$, train-time logit adjustment selects half the prior correction, $\tau = 0.5$, in 17 of 30 shards; its post-hoc form selects this value in 20 of 30. The full correction $\tau = 1$ is selected in 9 and 10 shards, respectively, and $\tau = 0.25$ in 4 and none. CUDA selects its lowest augmentation probability, $p_{\text{aug}} = 0.25$, in 17 of 30 shards and never selects $p_{\text{aug}} = 1$. On the balanced allocation, it selects 0.25 in 20 of 30 shards. End-to-end GCL selects $\sigma = 0.05$ in 18 of 30 shards; its classifier-only form selects the lowest value, $\sigma = 0.02$, in 29. DisAlign selects $\gamma = 1.5$ in 19 shards, and balanced sampling selects full balancing ($s = 1$) in 19. Selections at $\rho = 50$ resemble those at $\rho = 100$, except that DisAlign splits evenly between $\gamma = 1.5$ and its largest value, $\gamma = 2$ (14 shards each). At native shares, DisAlign selects its largest value, $\gamma = 3$, in 16 of 30 shards. Train-time logit adjustment selects all three values (9, 8, and 13 shards).

4.7 Probability quality

Imbalance does not worsen the LoRA model's raw probabilities, which are overconfident at every allocation. Balanced CE has a raw NLL of 1.37 nats and a raw ECE of 16.8 points; at $\rho = 100$, these values are 1.32 nats and 16.0 points. Temperature scaling removes most of this miscalibration and reveals a small cost of imbalance: scaled NLL rises from 1.03 to 1.07 nats and scaled ECE from 2.3 to 3.4 points. At $\rho = 100$, CUDA has by far the lowest raw NLL, 1.08 nats [1.02, 1.15], and raw ECE, 11.5 points [10.5, 12.6]. cRT (1.48 nats) and balanced sampling (1.43 nats) have the highest raw NLL. After scaling, NLL ranges from 0.98 (CUDA) to 1.07 nats across methods. Scaled ECE ranges from 1.4 points (classifier-only GCL) to 2.7 points

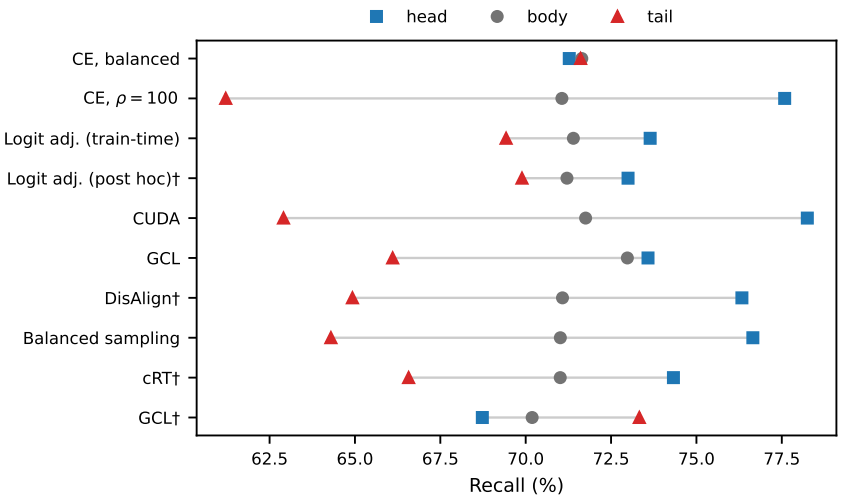


Figure 2: Logit adjustment and classifier-only Gaussian clouded logit (GCL) adjustment bring tail recall back to the balanced level at $\rho = 100$; CUDA (curriculum of data augmentation) raises recall in all three groups and retains the spread of cross-entropy (CE). Patient-macro recall of the head, body, and tail thirds of classes by assigned rank, averaged over shards. cRT: classifier re-training; DisAlign: distribution alignment. †Classifier only.

(balanced sampling and DisAlign), versus 3.4 for CE. The scaled intervals are unpaired and overlap. CUDA's augmentation curbs raw overconfidence, but temperature scaling brings every other method close to it.

5 Discussion

How much does each method recover? On TCGA-UT, logit adjustment closes the gap. At $\rho = 100$, both forms recover the full 1.56 points within the uncertainty of the paired recovery share and restore tail recall from 61 to almost 70 percent. No other method recovers the full damage: CUDA, both forms of GCL, DisAlign, balanced sampling, and cRT recover between two-fifths and two-thirds of it. At $\rho = 50$, all differences between methods remain below the one-point practical threshold. At native shares, differences reach 1.46 points between CUDA and classifier-only GCL, despite damage of only 0.42 points.

Is CUDA's gain an imbalance correction? No. CUDA improves the balanced model by 0.81 points, comparable to its recoveries of 0.84, 1.02, and 1.01 points at $\rho = 50$, $\rho = 100$, and native shares. Its imbalance-specific recovery is indistinguishable from zero at every allocation, and it leaves the gap between head and tail recall as wide as under CE. It is nevertheless useful: at native shares, it gives the most accurate model among the imbalanced allocations, at 72.10 percent BA, and it has the best raw probabilities at every imbalanced allocation. CUDA generally improves the LoRA model and could in principle be combined with a prior correction; this study did not test that combination.

Which conclusions carry over from BRACS? Four do. Logit adjustment and classifier-only GCL are the only methods that substantially restore tail recall. Post-hoc logit adjustment, a fixed prior subtraction at inference, matches its train-time form. CUDA's gain is mostly or entirely general. GCL and cRT make the balanced model slightly worse and suffer less imbalance damage than their recovery against CE suggests. Three conclusions do not carry over. On BRACS, no method closed the gap, and logit adjustment recovered about two-fifths of the 4.95-point damage; on TCGA-UT, it recovers all of the 1.56-point damage. CUDA gave the most accurate model at $\rho = 100$ on BRACS but ranks third here, because its general gain

is smaller than logit adjustment's imbalance-specific recovery. Classifier-only GCL, which gave the largest imbalance-specific recovery on BRACS, recovers 1.40 points here, about as much as end-to-end GCL (1.48) and less than train-time logit adjustment (1.54). Validation also selects a weaker prior correction, $\tau = 0.5$ instead of $\tau = 1$. These comparisons are descriptive because the datasets differ in class count, patient support, and training length.

Why can the gap be closed here? This study does not test the explanation, but the result fits the frozen-readout analysis of TCGA-UT (Queisler, 2026a). There, the frequency effect alone was the largest single source of damage. Limited tail support cost little by itself and mainly amplified the frequency effect. A method that removes the frequency effect then also removes most of the interaction. Under LoRA, the damage is smaller to begin with, and the tail retains 61 percent recall at $\rho = 100$, so the model has learned the rare classes and mainly misplaces their decision boundary. On BRACS, tail recall fell to 5 percent, which a boundary shift cannot repair.

Does LoRA itself help? Partly. Under LoRA, the damage at $\rho = 100$ is 1.56 instead of 2.91 points. Of this 1.35-point difference, 0.55 points come from a less accurate balanced model and 0.81 from a more accurate imbalanced model, mostly through body and tail recall. The best mitigated model at $\rho = 100$, train-time logit adjustment at 71.49 percent BA, remains 0.57 points below the frozen balanced readout. Adaptation therefore does not raise the accuracy available on balanced data, but it makes the encoder less sensitive to imbalance on TCGA-UT, which it did not on BRACS.

Limitations. The study covers one dataset, one encoder, and one adaptation scheme. The selected training length is the largest candidate tested. Validation accuracy still rose slowly between 1,200 and 2,400 steps, from 69.1 to 69.8 percent, so the LoRA models may not have converged. Longer training could change the balanced reference and thus every damage estimate. Four selections concentrate at unextended grid edges: CUDA's lowest augmentation probability at every allocation; DisAlign's largest γ at native shares and, in part, at $\rho = 50$; and the lowest σ of GCL's classifier-only form at $\rho = 100$. A wider grid could raise these three methods' recoveries, so their values are lower bounds on what the methods can reach. Damage at $\rho = 50$ and native shares is below one point, making the recovery shares unstable and the one-point criterion uninformative. Imbalance-specific recovery assumes that a method's effect on the balanced allocation carries over additively to imbalanced allocations; the design does not test this assumption. Comparisons with the frozen readout and BRACS are descriptive and have no paired intervals.

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